



HIGHER DIPLOMA IN SOFTWARE ENGINEERING 25.2F

SMART HEALTH MONITORING WEARABLE WITH REAL- TIME VITAL ANALYSIS

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Abstract

- ✚ This report details the design and implementation of a low-cost, smart health monitoring wearable device, a critical innovation for Sri Lanka's healthcare sector. The system utilizes an ESP32 microcontroller integrated with medical-grade sensors—a MAX30102 for heart rate and blood oxygen saturation (SpO₂), an AD8232 for electrocardiogram (ECG) readings, and a for non-contact skin temperature measurement. Vital signs are displayed in real-time on a 2.4" TFT screen, and the device provides immediate audio alerts via a DF Player Mini upon detecting abnormal health conditions, such as fever, hypoxia, or cardiac arrhythmia.
- ✚ Our approach prioritizes affordability, power efficiency, and localized calibration for tropical climates. The prototype demonstrates how digital logic and embedded systems can be pragmatically applied to create life-saving technology. This project exemplifies the potential of locally developed medical wearables to bridge healthcare gaps in rural and urban Sri Lankan communities, offering a proactive solution for personal health management and remote patient monitoring.

Key Points:

- Objective: To create an affordable, wearable device for real-time monitoring of critical health vitals to prevent medical emergencies.
 - Components: ESP32, MAX30102, AD8232, MLX90614 , 2.4" TFT Display, DF Player Mini, Arduino Mega, Arduino Uno
-

- **Methodology:** Sensor data acquisition and processing via ESP32, with logic-based analysis for abnormality detection, Arduino-Mega and Arduino-Uno Boards used to code the 2.4" TFT Display for the Smart Digital vital framework.
- **Benefits:** Low cost, proactive health alerts, and designed for Sri Lankan environmental conditions.
- **Future Considerations:** Integration with mobile health platforms and cloud-based data logging.

Acknowledgement

We acknowledge the support and guidance provided to us by our lecturers, instructors, peers, and others who contributed to the successful completion of this project.

- ✚ **Instructors and Mentors:** Our project would not have been possible without the expert guidance and encouragement from our course instructors. Their deep knowledge of digital systems, microcontroller programming, and practical electronics was invaluable in shaping this project from concept to reality.
 - ✚ **Peers and Collaborators:** We are grateful for the collaborative environment fostered by our peers. Their feedback during brainstorming sessions and their technical insights during the debugging phase were crucial to our problem-solving process.
 - ✚ **Friends and Family:** We extend our heartfelt gratitude to our friends and family members for their unwavering support and patience throughout the duration of this project.
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Project Architecture

PIP HEALTH 3000 — Wearable Health Monitor Enclosure

Retro Pip-Boy inspired body with ESP-32 + sensors

Global & toggles

\$fn = 100; // smoothness

SHOW_EXPLODED = false; // true => lifts backplate off for inspection

EXPLODE_Z = 6; // lift amount if exploded

WALL = 2; // default wall thickness

TOL = 0.4; // fit tolerance for cavities & covers

Main body dimensions

device_len = 110.5;

device_wid = 72;

device_ht = 18; // total enclosure height (without knobs)

corner_rad = 8;

Wrist band / hinge mount

band_w = 25;

band_thk = 3;

hinge_block = [22, 10, 10]; // [x,y,z] each end

hinge_pin_d = 4.2; // through pin/rod diameter

hinge_ear = 3; // ear thickness

Component envelopes (inner cavities)

ESP-32 devkit (typical narrow)

```
esp32_len = 28; esp32_wid = 18; esp32_ht = 8;
```

```
esp32_pos = [14, device_wid - 27, 0];
```

Arduino Mega 2560

```
mega_len = 102; mega_wid = 53; mega_ht = 15;
```

```
mega_pos = [(device_len - mega_len)/2, (device_wid - mega_wid)/2, 0];
```

TFT Display window & recess (front)

```
disp_w = 60; disp_h = 40; disp_thk = 4;
```

```
disp_pos = [(device_len - disp_w)/2, (device_wid - disp_h)/2, 0];
```

DFPlayer Mini

```
dfp_len=16; dfp_wid=22; dfp_ht=6;
```

```
dfp_pos=[device_len - 30, 10, 0];
```

Speaker (round)

```
spk_d = 36; spk_ht = 6;
```

```
spk_pos=[30, device_wid/2, 0];
```

MAX30102 (SpO2/HR)

```
max_len=10; max_wid=16.5; max_ht=5;
```

```
max_pos=[10, (device_wid - max_wid)/2, 0];
```

AD8232 ECG module (similar small board)

```
ad_len=10; ad_wid=12; ad_ht=5;
```

```
ad_pos=[device_len - 25, device_wid - 20, 0];
```

MLX90614 IR thermometer (round window)

```
mlx_d=10; mlx_ht=6;
```

```
mlx_pos=[device_len - 15, device_wid/2, 0];
```

Battery compartment (internal)

```
bat_len=60; bat_wid=30; bat_ht=10;
```

```
bat_pos=[(device_len - bat_len)/2, (device_wid - bat_wid)/2, 0];
```

Helpers

```
module rounded_box(x,y,z,r) {
```

```
    constant-wall rounded box (solid)
```

```
    hull() {
```

```
        for(a=[0,1], b=[0,1])
```

```
            translate([ a?(x-r):r, b?(y-r):r, 0 ]) cylinder(h=z, r=r);
```

```
    }
```

```
}
```

```
module shell_rounded(x,y,z,r, t=WALL) {
```

```
    solid shell with inner cavity
```

```
    difference() {
```

```
        rounded_box(x,y,z,r);
```

```
        translate([t, t, t])
```

```
            rounded_box(x-2*t, y-2*t, z - t, max(0,r-t));
```

```
    }
```

```
}
```

```
module standoff(h=6, od=6, id=2.5) {
```

```
difference() {
    cylinder(h=h, d=od);
    translate([0,0,-0.1]) cylinder(h=h+0.2, d=id);
}
}

module grill(x0,y0, nx=5, ny=3, dx=15, dy=20, d=2.2) {
    through holes for ventilation
    for(i=[0:nx-1])
        for(j=[0:ny-1])
            translate([x0+i*dx, y0+j*dy, -0.1]) cylinder(h=device_ht+1, d=d);
}

module counterbore_through(d_thru=2.4, d_bore=5, h_bore=2) {
    for screw holes (M2 self-tapping typical)
    union() {
        cylinder(h=device_ht+1, d=d_thru);
        translate([0,0,device_ht-h_bore])
            cylinder(h=h_bore+0.2, d=d_bore);
    }
}

module screen_bezel() {
    Raised retro bezel & lip around screen
    bezel_pad = 2; // frame around window
    lip      = 1.6; // rim height
    translate([disp_pos[0]-bezel_pad, disp_pos[1]-bezel_pad, device_ht - lip])
        difference() {
```



```

        cube([disp_w + 2*bezel_pad, disp_h + 2*bezel_pad, lip]);
        translate([bezel_pad, bezel_pad, -0.1])
            cube([disp_w, disp_h, lip+0.2]);
    }
}

module sensor_windows() {
    Openings for optical/IR sensors + speaker grill cut
    Speaker through grill
    translate([spk_pos[0], spk_pos[1], -0.1]) cylinder(h=device_ht+2, d=spk_d - 6);

    MLX90614 circular window
    translate([mlx_pos[0], mlx_pos[1], -0.1]) cylinder(h=device_ht+2, d=mlx_d - 2);

    MAX30102 rectangular window
    translate([max_pos[0], max_pos[1], -0.1])
        cube([max_len, max_wid, device_ht+2]);
}

module side_knobs() {
    Two retro control knobs on right edge
    knob_h = 10; knob_d = 12;
    translate([device_len + 1, device_wid*0.35, device_ht/2 - knob_h/2])
        cylinder(h=knob_h, d=knob_d);
    translate([device_len + 1, device_wid*0.65, device_ht/2 - knob_h/2])
        cylinder(h=knob_h, d=9);
}

```

```

module strap_hinge(left=true) {
    Rugged hinge block with through pin hole and strap slot
    blk = hinge_block;
    position at center along width
    translate([ (device_len - blk[0])/2,
                left ? -blk[1] : device_wid,
                0 ])
    difference() {
        outer block
        cube(blk);
        pin hole
        translate([blk[0]/2, blk[1]/2, blk[2]/2])
            rotate([90,0,0]) cylinder(h=blk[1]+0.4, d=hinge_pin_d);
        strap slot (for a loop strap)
        translate([(blk[0]-band_w)/2, left?(-0.1):(blk[1]-band_thk+0.1), blk[2]/2 - band_thk/2])
            cube([band_w, band_thk+0.2, band_thk]);
        lighten
        translate([2,2,2]) cube([blk[0]-4, blk[1]-4, blk[2]-4]);
    }
}

module label_engrave() {
    translate([device_len/2, device_wid + 6, device_ht/2 - 0.5])
    linear_extrude(1)
    text("PIP HEALTH 3000", size=6, halign="center", font="Liberation Sans:style=Bold");
}

```

Main body & internals

```
module body_top() {  
    main shell with wall + front openings + grills + engraving  
    difference() {  
        shell_rounded(device_len, device_wid, device_ht, corner_rad, WALL);  
  
        Display cavity (recess, not full through)  
        translate([disp_pos[0], disp_pos[1], device_ht - (disp_thk + 2)])  
            cube([disp_w, disp_h, disp_thk + 2 + 0.01]);  
  
        Component cavities (open downward from inside)  
        translate([esp32_pos[0], esp32_pos[1], WALL - 0.1])  
            cube([esp32_len+TOL, esp32_wid+TOL, esp32_ht + 3]);  
        translate([mega_pos[0], mega_pos[1], WALL - 0.1])  
            cube([mega_len+TOL, mega_wid+TOL, mega_ht + 3]);  
        translate([dfp_pos[0], dfp_pos[1], WALL - 0.1])  
            cube([dfp_len+TOL, dfp_wid+TOL, dfp_ht + 3]);  
        translate([bat_pos[0], bat_pos[1], WALL - 0.1])  
            cube([bat_len+TOL, bat_wid+TOL, bat_ht + 4]);  
  
        Speaker / sensor through windows  
        sensor_windows();  
  
        Vent grills  
        grill(20, 15, 5, 3, 15, 20, 2.2);  
  
        Screw holes (downward through for backplate screws)  
        for (p = [
```

```
[bat_pos[0]+5,          bat_pos[1]+5],  
[bat_pos[0]+bat_len-5,   bat_pos[1]+5],  
[bat_pos[0]+5,          bat_pos[1]+bat_wid-5],  
[bat_pos[0]+bat_len-5,   bat_pos[1]+bat_wid-5]  
) translate([p[0], p[1], -0.1]) counterbore_through();
```

Front face engraving recess

```
label_engrave();  
}
```

Standoffs around battery bay for backplate screws

```
for (p = [  
    [bat_pos[0]+5,          bat_pos[1]+5],  
    [bat_pos[0]+bat_len-5,   bat_pos[1]+5],  
    [bat_pos[0]+5,          bat_pos[1]+bat_wid-5],  
    [bat_pos[0]+bat_len-5,   bat_pos[1]+bat_wid-5]  
) translate([p[0], p[1], WALL])  
    standoff(h=6, od=6, id=1.8);
```

Bezel & side details

```
screen_bezel();  
side_knobs();
```

Strap hinge blocks

```
strap_hinge(true);  
strap_hinge(false);  
}
```

```
module backplate() {  
    simple rounded rectangle plate that mates to the shell, with screw clearance holes  
  
    plate_t = 2;  
    explode = SHOW_EXPLODED ? EXPLODE_Z : 0;  
  
    translate([0,0,-plate_t - explode])  
    difference() {  
        outer plate slightly inset to fit into shell lip  
        translate([WALL, WALL, 0])  
        rounded_box(device_len-2*WALL, device_wid-2*WALL, plate_t, max(0,corner_rad -  
WALL));  
        screw clearance  
        for (p = [  
            [bat_pos[0]+5,          bat_pos[1]+5],  
            [bat_pos[0]+bat_len-5,    bat_pos[1]+5],  
            [bat_pos[0]+5,          bat_pos[1]+bat_wid-5],  
            [bat_pos[0]+bat_len-5,    bat_pos[1]+bat_wid-5]  
        ]) translate([p[0], p[1], -0.1]) cylinder(h=plate_t+0.3, d=2.6);  
        lightening pocket under speaker  
        translate([spk_pos[0]-spk_d/2+3, spk_pos[1]-spk_d/2+3, 0.4])  
            cylinder(h=plate_t, d=spk_d-6);  
    }  
  
    subtle ribs for stiffness  
    rib_w = 3; rib_h = 1.2;  
    for(x=[WALL+10 : 16 : device_len-WALL-10])  
        translate([x, device_wid*0.25, -rib_h - explode]) cube([rib_w, device_wid*0.5, rib_h]);  
}
```

Assembly

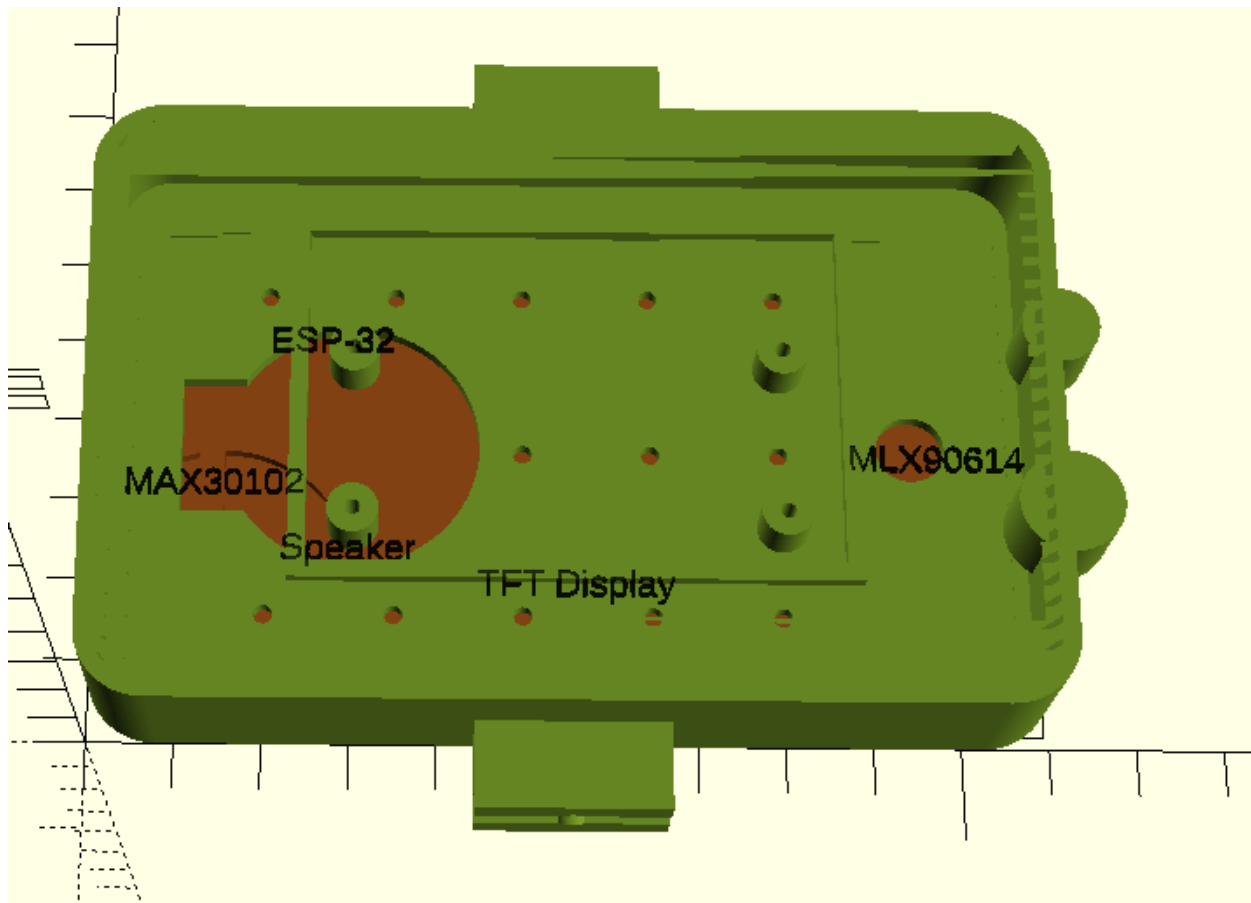
```
module assembly() {  
    color("olivedrab") body_top();  
    color("saddlebrown") backplate();
```

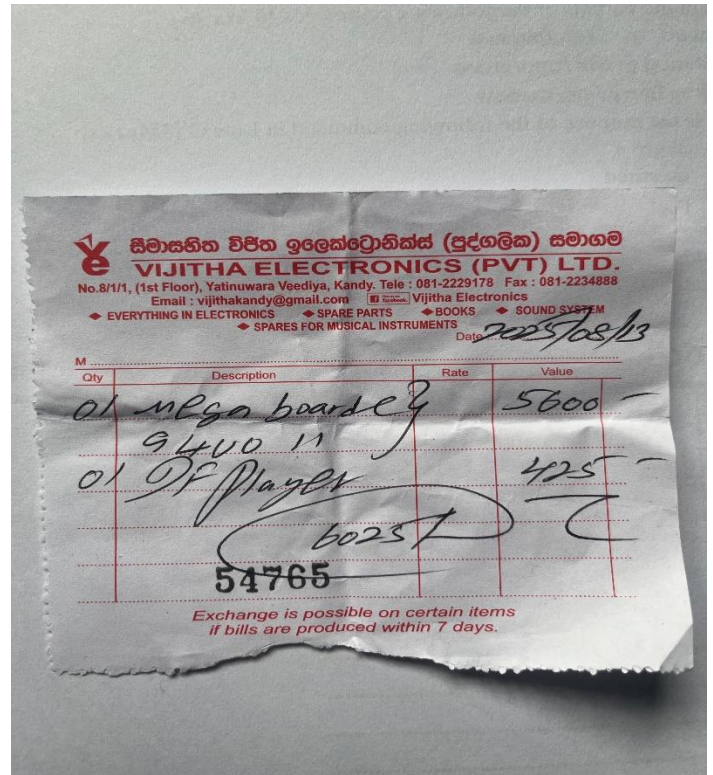
Optional labels near cavities (non-printing emboss, very shallow)

```
color([0,0,0,0.7]) {  
    translate([disp_pos[0]+disp_w/2, disp_pos[1]-4, device_ht-0.6])  
        linear_extrude(0.8) text("TFT Display", size=3, halign="center");  
    translate([spk_pos[0], spk_pos[1]-20, device_ht-0.6])  
        linear_extrude(0.8) text("Speaker", size=3, halign="center");  
    translate([max_pos[0]+max_len/2, max_pos[1]-4, device_ht-0.6])  
        linear_extrude(0.8) text("MAX30102", size=3, halign="center");  
    translate([mlx_pos[0], mlx_pos[1]-9, device_ht-0.6])  
        linear_extrude(0.8) text("MLX90614", size=3, halign="center");  
    translate([esp32_pos[0]+esp32_len/2, esp32_pos[1]-4, device_ht-0.6])  
        linear_extrude(0.8) text("ESP-32", size=3, halign="center");  
}  
}
```

Render

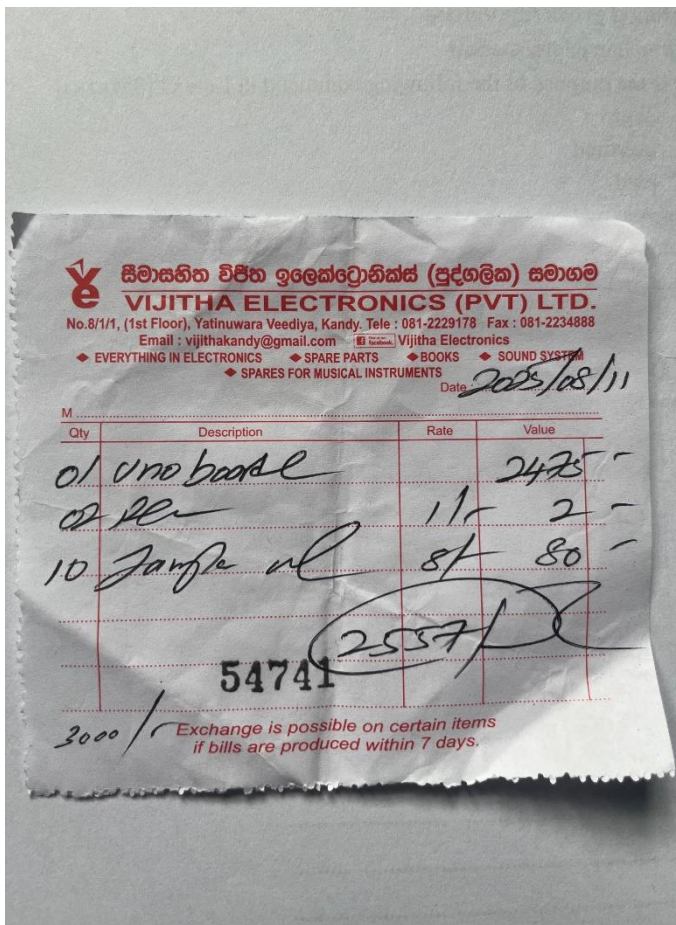
```
assembly();
```





Budget Review

3d Printing Enclosure: - Rs.5400/=



Data Set Review

Smart Health-Care Wearable Device – Project - NIBM

Vital Sign	Healthy Range	Moderate Risk (Monitor)	High Risk (Alert)	Source
Heart Rate (bpm)	60 - 100	50-59 or 101-119	< 50 or > 120	Sri Lanka College of Physicians
SpO ₂ (%)	≥ 95	91 - 94	≤ 90	WHO Guidelines
Temperature (°C)	36.5 - 37.5	37.6 - 38.3	> 38.3	Ministry of Health, SL

"This letter is to confirm that I, Dr. Prabha Nanayakkara, have reviewed the proposed clinical parameters and decision-making algorithms for the smart health monitoring wearable device developed by Chamod Minidu in the NIBM Institute Kandy.

The thresholds for vital signs such as heart rate, blood oxygen saturation (SpO₂), and body temperature, as presented to me, are in line with standard clinical practice and guidelines used in Sri Lanka for initial patient assessment and early warning of potential health deterioration.

The concept of using these parameters to trigger alerts in a wearable device is a sound and medically relevant approach for promoting proactive health awareness.

I wish the student success in this valuable project."

Name

Dr. Prabha Wasanthi Nanayakkara,

Signature Stamp

Dr. P.W. Nanayakkara

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Date

21/08/2025